

ADiDL: Improving Difference-in-Differences (DiD) Estimators by Incorporating Dynamics

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1 Introduction

Stationarity is one of the most fundamental concepts in econometrics. This is because a researcher is likely to find a spurious relationship between two variables that are observed over time if they are nonstationary (Granger and Newbold [1974](#)). The problem is that neglected nonstationarity in the outcome induces nonstationarity in the error term, leading to overconfidence and bias due to now common trending in both the independent variable and error term.¹ The recommended best practice is to not neglect the nonstationarity, with researchers being required to difference both variables, look for a cointegrating relationship between them, or include lags of both variables in their model (see Kagalwala and Whitten [Forthcoming](#)).

As the literature on dealing with nonstationary data has developed, so has the so-called credibility revolution in the social sciences. Most researchers evaluating relationships between variables observed over time now employ a differences-in-differences (DiD) design if the variables can also be observed in different units. Concerns about temporal processes in the error term have independently arose in this literature as well, with the recommended best practice being to estimate clustered standard errors. In an extremely influential paper,² Bertrand, Duflo, and Mullainathan ([2004](#), p. 270) claim that clustered standard errors are “consistent in the presence of *any* correlation pattern within states over time” (emphasis

¹The resulting bias is actually more complicated, having to do with the fact that the variables are non-ergodic and have a joint distribution with non-constant sample moments (see Phillips [1986](#)).

²Google Scholar reports over 15,000 cites for this article.

original).³ Again, MacKinnon, Nielsen, and Webb (2023, p. 273) state that “Any pattern of heteroskedasticity and/or dependence is allowed within each cluster.” Clearly, this implies that the error term may even be nonstationary within units.

This advice has arguably led to the prevailing practice of using design-based estimators, like DiD, and ignoring concerns about dynamics or stationarity so long as clustered standard errors are also estimated. However, given the literature on addressing nonstationarity discussed above, it is not obvious that estimating clustered standard errors is the best solution when executing a design-based estimator in the presence of nonstationarity. In this paper, I show that including lags of both the outcome and the post-treatment dummy variable in a DiD, TWFE regression improves the performance of the estimator when there is neglected nonstationarity more than estimating clustered standard errors does.⁴ This modeling strategy is analogous to specifying the ADL(1,1) model that is commonly used in time-series and shown to perform well even when data are nonstationary. As a result, I dub this model the ADiDL(1,1).

To be clear, the ADiDL(1,1) with standard errors estimated *iid* performs better when data are nonstationary than a static DiD regression with clustered standard errors even under conditions most favorable to the latter model: homogeneous treatment effects, identical treatment timing, and a large number of clusters ($N = 1000$). While static DiD regressions appear to avoid the more serious problem of bias that is induced from nonstationary data,⁵ they suffer considerably from overconfidence and poor precision. Moreover, even under conditions not favorable to DiD, TWFE regression in general (i.e., heterogeneous treatment effects and staggered treatment timing), the ADiDL(1,1) still recovers the true average treatment effect and does so at any level of autocorrelation in the data. As a result, the ADiDL(1,1) should be the preferred design-based model when data are nonstationary and should be

³The promise of clustered standard errors lies in the assumption that error terms across units are independent. The consistency of clustered standard errors as the number of units grows asymptotically is not disputed in this article.

⁴Replication files can be accessed at [Replication Files](#).

⁵This is likely because the independent variable, the post-treatment dummy, cannot be nonstationary. Thus, common stochastic trending cannot occur between the error term and independent variable.

especially preferred when treatment effects are heterogeneous and/or applied at different times.

Section 2 introduces the ADiDL(1,1) model and presents Monte Carlo results that support the first claim that ADiDL(1,1) performs better than a static DiD with clustered standard errors when there is neglected nonstationarity. Section 3 presents the results from simulations that assess how the ADiDL(1,1) performs in the presence of heterogeneous effects, staggered treatment timing, and heterogeneous effects with staggered timing. The results in section 3 support the argument that the TWFE estimator performs well in the ADiDL(1,1) model despite these complications that are otherwise known to cause issues in static TWFE estimates.

2 Clustered Standard Errors vs. the ADiDL(1,1)

The conventional DiD regression is often written as (e.g., Goodman-Bacon [2021](#), p. 255):

$$y_{it} = \gamma_i + \gamma_t + \beta D_{it} + e_{it}, \quad (1)$$

where y_{it} is an outcome of interest observed in units $i \in \{1, \dots, N\}$ over periods $t \in \{1, \dots, T\}$, γ_i and γ_t are fixed effects or dummy variables for each unit and period, respectively, and D_{it} is a dummy variable that equals 0 for all periods before the treatment period t' and 1 for $t \geq t'$ in units that receive treatment. Under the assumption that the error term e is *iid* across units (i.e., $e_i \perp e_j \forall$ units $i \neq j$), estimating clustered standard errors generates consistent estimates of $sd(\hat{\beta})$ as the number of units or clusters grows even if there is unmodeled dependence within units. This model is the prevailing standard and will be referred to as the static DiD with clustered standard errors. Of course, this model is motivated under the parallel trends assumption that “the average outcomes for treated and comparison groups would have followed parallel paths over time” in the absence of treatment (Callaway and Sant’Anna [2021](#), p. 200). When this assumption holds, the average effect of the treatment

on the units that receive it is identified (i.e., $\mathbb{E}[\hat{\beta}] = \beta$) and reduces to the average treatment effect.

The ADiDL(1,1) model is written as:

$$y_{it} = \alpha y_{it-1} + \gamma_i + \gamma_t + \beta_1 D_{it} + \beta_2 D_{it-1} + e_{it}, \quad (2)$$

where y_{it-1} is the lag of the outcome and D_{it-1} is the lag of the post-treatment dummy variable. Instead of a post-regression correction of the standard errors, like clustering, the ADiDL(1,1) models the dependence in the error term directly. For example, if there truly was neglected temporal dependence or autocorrelation in the outcome the error term from the static DiD would contain αy_{it-1} . Explicitly including αy_{it-1} in the DiD regression models this dependence rather than corrects for it post-estimation. Including $\beta_2 D_{it-1}$ also helps model any additional temporal dependence in the error term. Equation (2) estimated with *iid* standard errors will continue to be referred to as the ADiDL(1,1) and is the main alternative I will consider to the static DiD with clustered standard errors.

Figure 1 compares the performance of the static DiD regression with clustered standard errors to the ADiDL(1,1) when the outcome is autocorrelated but stationary ($\alpha = 0.5$). As expected, static DiD with clustered standard errors does an excellent job in this scenario: the estimates of the short-run treatment effect, $\hat{\beta}$, are unbiased while the estimates of $sd(\hat{\beta})$ are also unbiased and weakly smaller than those from the ADiDL(1,1). The static DiD with clustered standard errors also exhibits higher power. The results shown in Figure 1 are from a Monte Carlo experiment with a reasonable amount of clusters ($N = 50$) and time periods ($T = 10$) for a DiD design. The true treatment effect at $t' = 5$ was 0 to assess the presence of spuriousness.

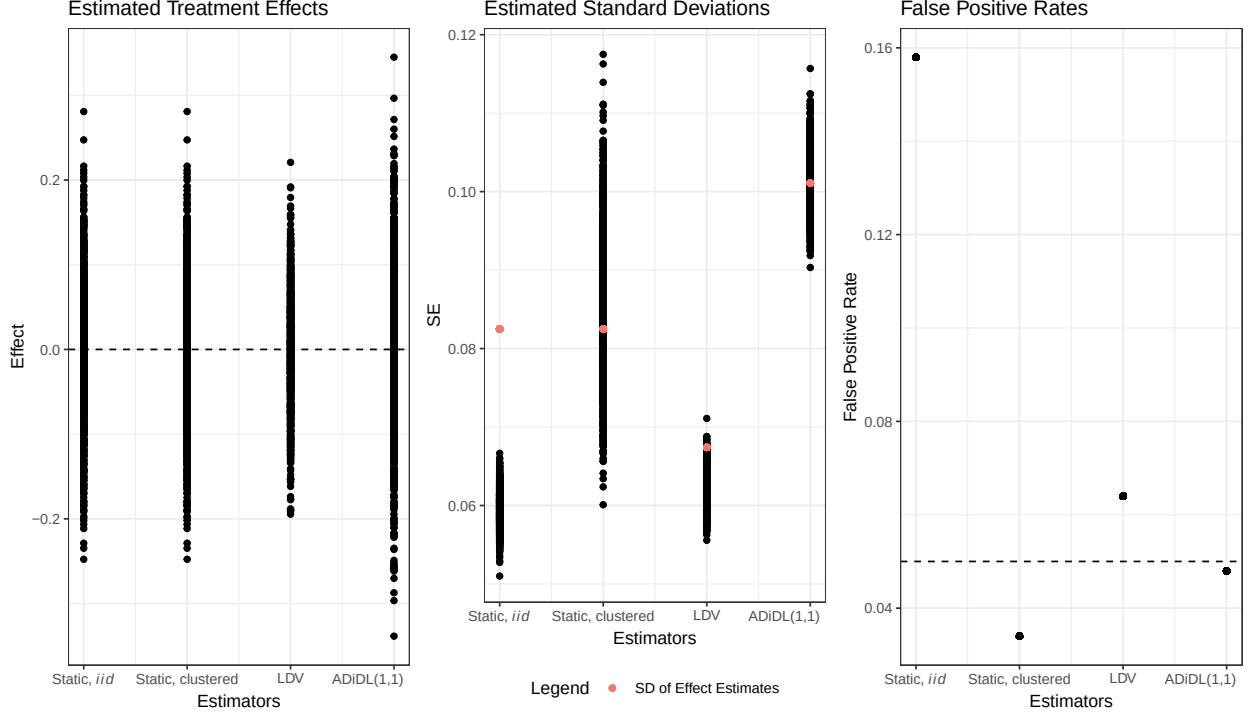


Figure 1: Comparing the performance of the static DiD with clustered standard errors to the ADiDL(1,1) when the outcome is autocorrelated but stationary ($\alpha = 0.5$). A static DiD with standard errors estimated *iid* and a lagged dependent variable (LDV) DiD model with *iid* standard errors is included for comparison. I conducted 1000 simulations where $\frac{N}{2}$ units were randomly selected to receive treatment and a burn-in of 100 periods was implemented.

However, this story is reversed when the outcome is a unit root. Figure 2 shows the results from the same experiment but when $\alpha = 1$. While all models again recover unbiased short-run treatment effects, the ADiDL(1,1) estimates standard errors that are unbiased and almost half the size of the clustered standard errors. The ADiDL(1,1) now also exhibits higher power. Both of these results hold even when the number of clusters is large ($N = 1000$).⁶ Figure 3 shows that the ADiDL(1,1)'s performance gains are even larger once the outcome becomes even slightly explosive ($\alpha = 1.05$).

⁶See Figure A.1 in the Appendix.

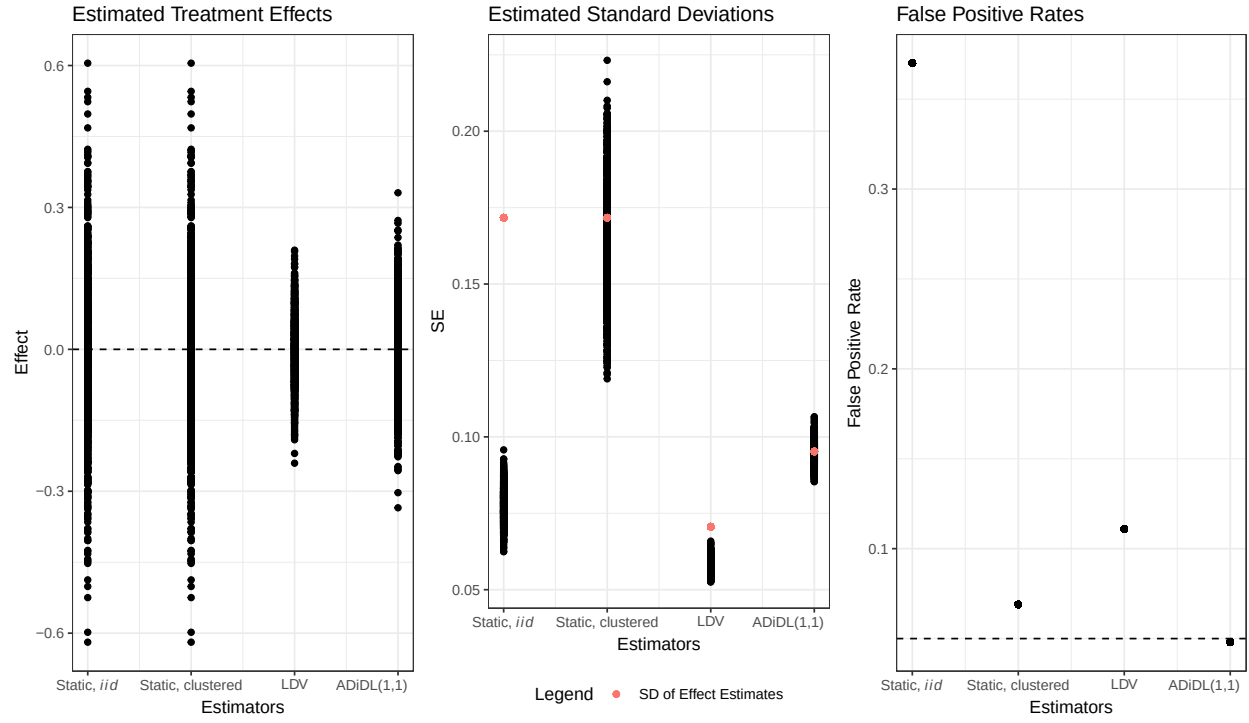


Figure 2: Comparing the performance of the static DiD with clustered standard errors to the ADiDL(1,1) when the outcome is a unit root ($\alpha = 1$).

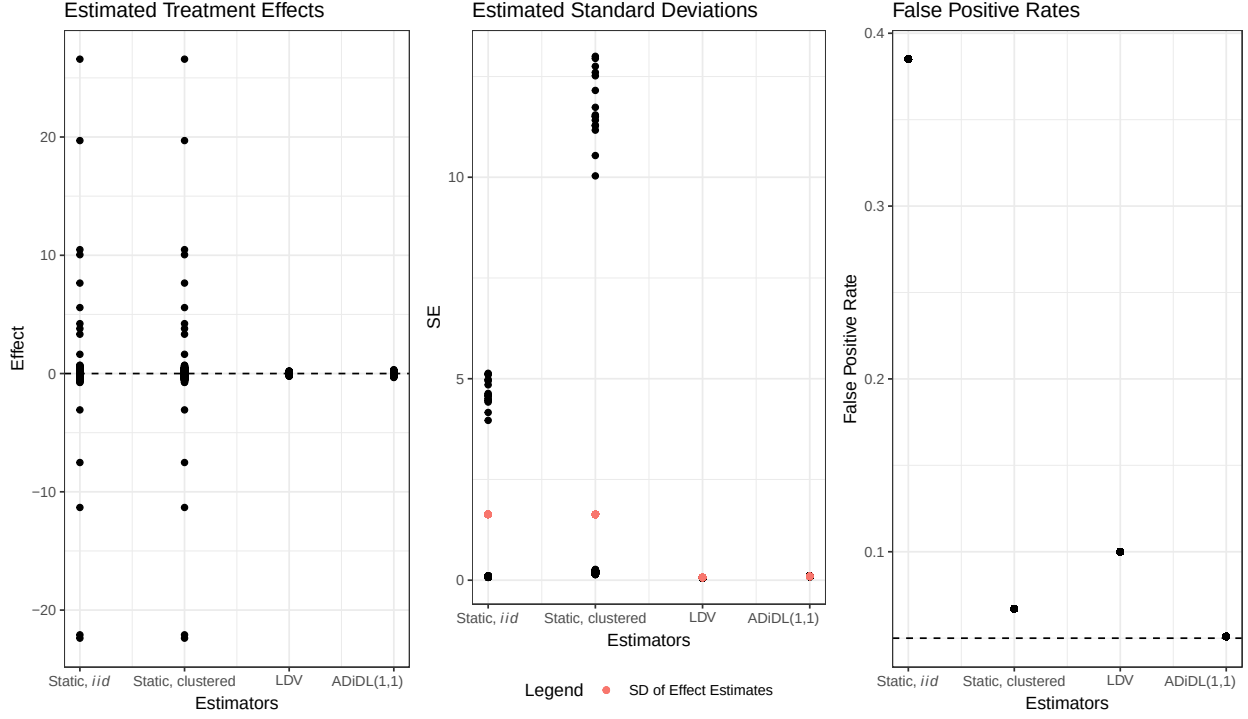


Figure 3: Comparing the performance of the static DiD with clustered standard errors to the ADiDL(1,1) when the outcome is explosive ($\alpha = 1.05$).

While these results may seem trivial, they suggest that the widespread reliance on clustered standard errors to perform well in the face of *any* unmodeled dynamics is misplaced. Instead, modeling the dynamics by including lags of the outcome and independent variable does more to improve DiD estimates than estimating clustered standard errors when the outcome is nonstationary.

3 The ADiDL(1,1) vs. Heterogeneous Effects and Staggered Timing

It is well known that the TWFE estimator poorly approximates a DiD design when there are heterogeneous treatment effects and/or staggered timing (Athey and Imbens 2022; Goodman-Bacon 2021; Sun and Abraham 2021; De Chaisemartin and d’Haultfoeuille 2020; Borusyak and Jaravel 2018). In this section I present the results from 3 experiments that innovate on the experiment reported in Figure 1. Figure 4 displays the results when there are heterogeneous effects, Figure 5 introduces staggered treatment timing, and Figure 6 includes both heterogeneous effects and staggered timing. Each experiment has the same general setup as the experiment explained for Figure 1 (i.e., 50 clusters or units, 10 time periods, 1000 simulations, and a burn-in period of 100 units). At the top of each figure is a graph of the average outcomes by treatment group using a simulated dataset that is representative of those generated during the 1000 runs per respective experiment. These average outcomes are intended to show what is being estimated and to show that there are indeed parallel trends before treatment occurs.⁷

The ADiDL(1,1) model recovers the true average treatment effect despite heterogeneous effects and staggered timing, but exhibits a relatively small amount of bias when both occur simultaneously. These results hold whether the outcome is stationary or not, but in all of the results presented in this section the outcome is a unit root. In all cases, as expected, the static DiD model demonstrates large amounts of bias in both the estimated average treatment effect and its estimated standard deviation.

⁷To ensure parallel trends, I generated N sequences of random noise that were correlated at 0.9. The treatment effects were then applied during a for-loop that also built in the autocorrelation. While this process may seem to violate the *iid* assumption on the error terms required for clustered standard errors to be consistent, all of the results in this paper hold when the random sequences are set to be completely uncorrelated. In that case, the DiD models are still identified due to random treatment assignment.

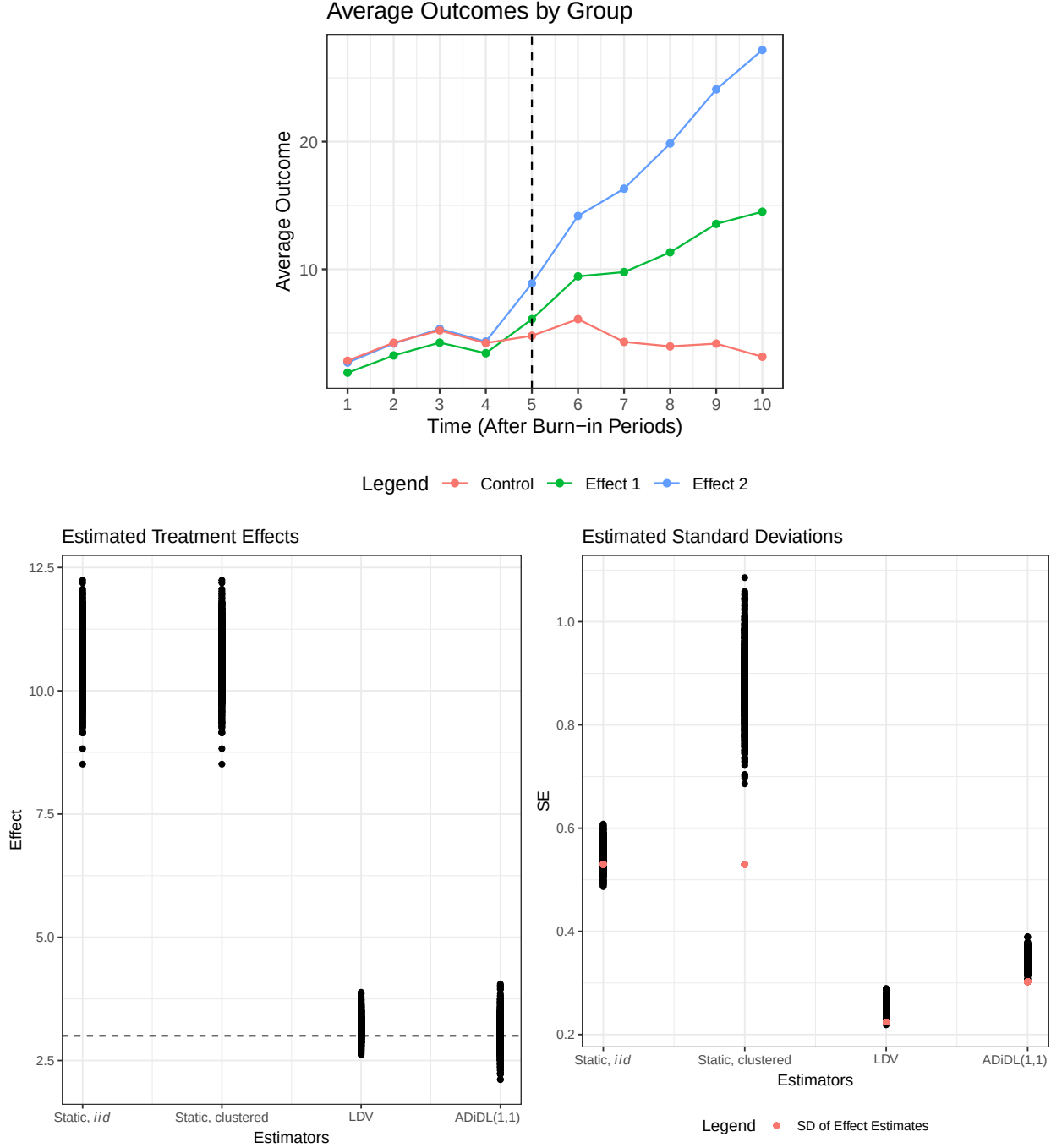


Figure 4: Assessing the performance of models used in the previous experiments when there are heterogeneous treatment effects. Specifically, treatment still occurs at $t' = 5$ but has an effect of 2 on half of the treated units and an effect of 4 on the other half of the treated units. Therefore, the true average treatment effect is 3. The ADiDL(1,1) model is the only model that appears unbiased, although its standard errors are biased in a conservative direction (i.e., underconfident).

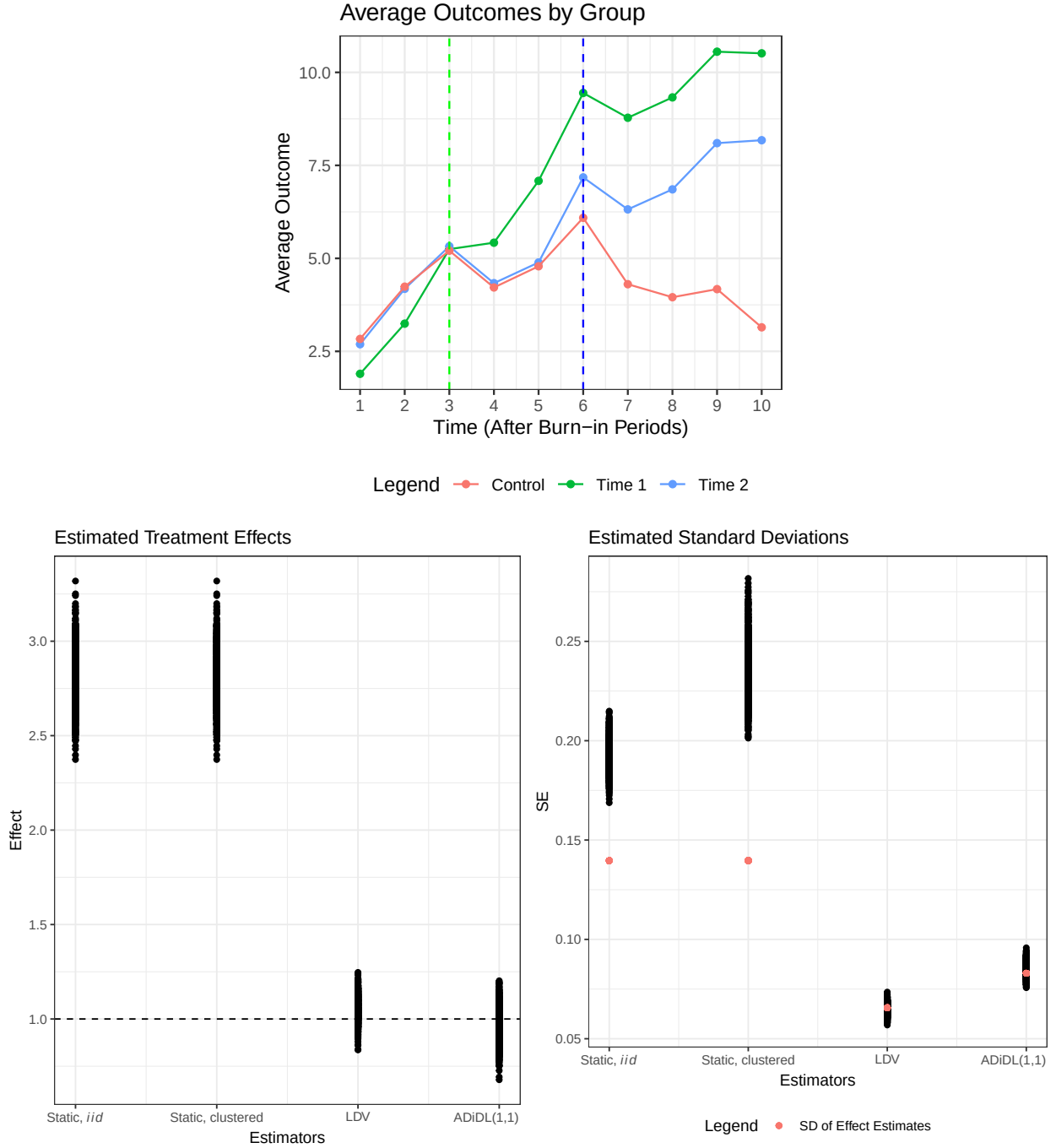


Figure 5: In this experiment, there is a single true effect of 1 but it is received by half of the treated group at $t' = 3$ and the other half at $t'' = 6$. Again, the TWFE estimator in the ADiDL(1,1) model performs well under these conditions.

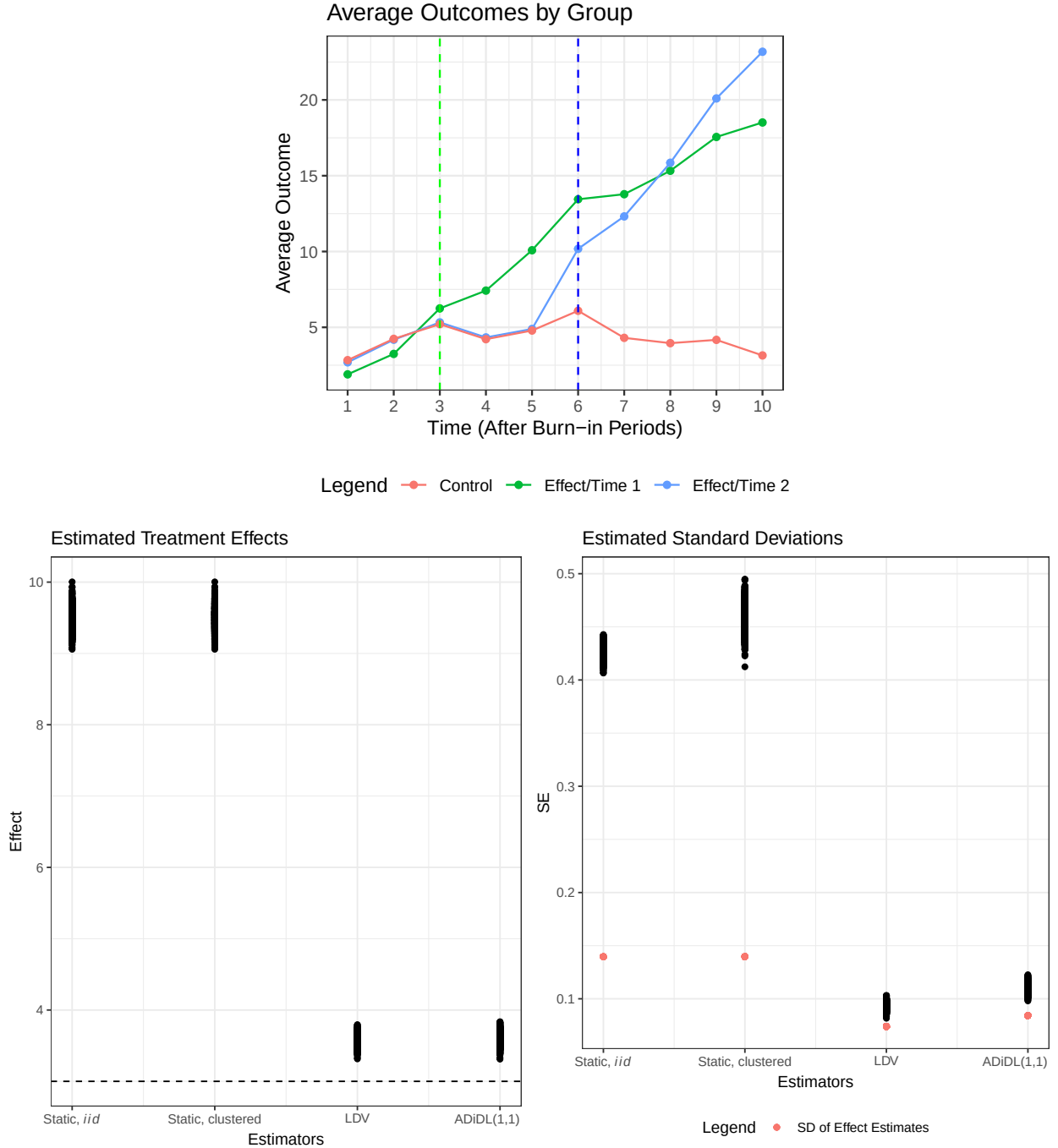


Figure 6: Finally, these results show the performance of the models under heterogeneous effects applied at different times. Specifically, half of the treated group receives a true effect of 2 at $t' = 3$ while the other half of the treated group gets a true effect of 4 at $t'' = 6$. In this case, the ADiDL(1,1) model is somewhat biased and underconfident. It appears to be biased upward by about 17% ($\frac{3.5-3}{3}$) while the static DiD models appear to exhibit an upward bias of about 217% ($\frac{9.5-3}{3}$).

4 Appendix

A Reevaluating Results in Figure 1 with 1000 Clusters

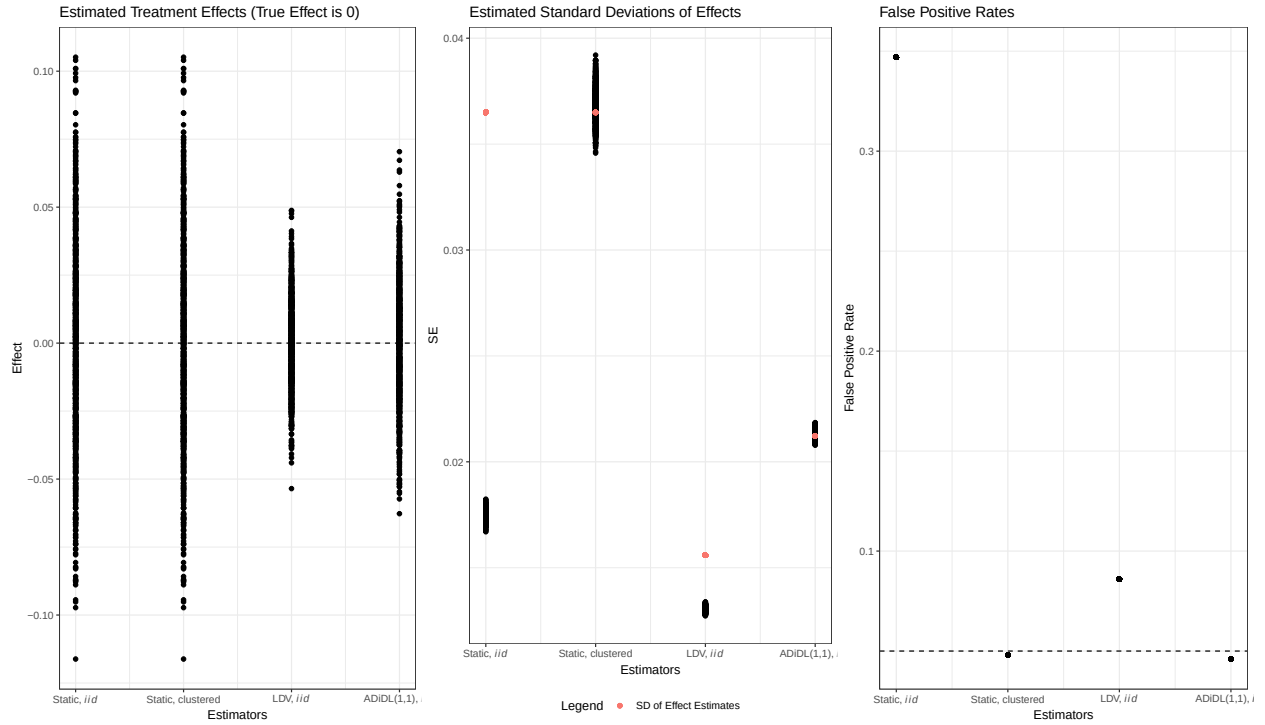


Figure A.1: This experiment is identical to that presented in Figure 1 except there are now 1000 units or clusters. Again, the ADiDL(1,1) has standard errors that are smaller and much more precise than the static DiD with clustered standard errors. It also has a slightly lower false positive rate, although the static model with clustered standard errors is more than appropriately powered.

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